Project description

1 Project title

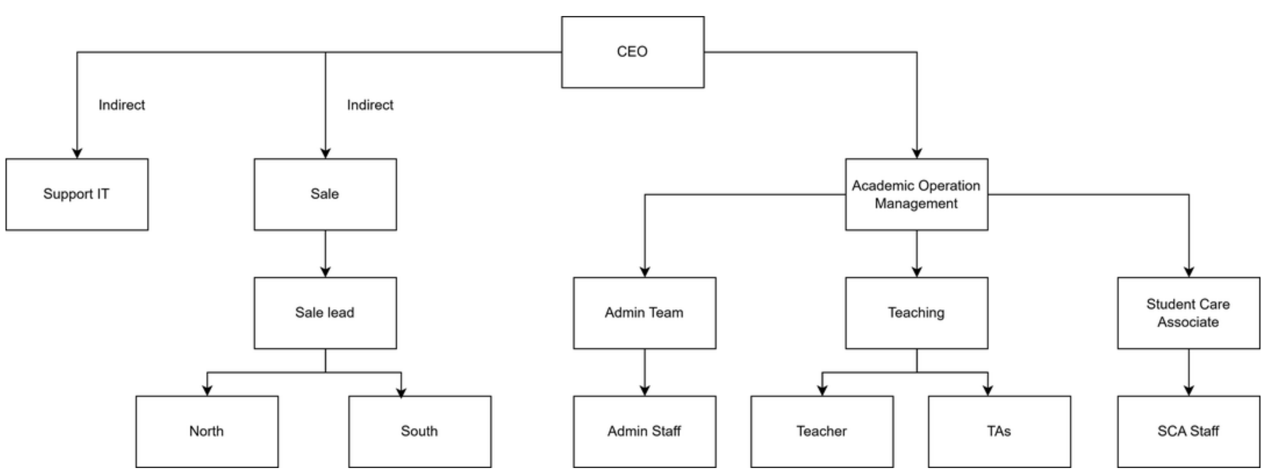
Integrated Learning Center Management System (ILCMS)

2 Customer description

The customer of this project is an online learning center that has been operating for approximately five years, since September 2021. The center currently serves around 300–500 students per year and operates about 15 active classes at any given time. Its teaching staff includes 45 teachers and 15–20 teaching assistants. The center provides only online learning, with all teaching, learning, and assessment activities conducted through digital platforms.

Currently, to manage daily operations, the center relies on a highly fragmented ecosystem of disconnected tools, including ECO, internal CMS, legacy LMS, ClassIn/ClassSeen, Zoom, Zalo/Facebook for communication, and scattered Excel spreadsheets. Because these systems do not communicate with each other, the center's staff faces critical operational bottlenecks:

- Financial tracking relies on manual screenshots sent via chat apps.
- Data synchronization is non-existent, forcing staff to manually type student information across multiple platforms.
- Management lacks real-time visibility into revenue, outstanding debts, and class capacities.



3 Business Process

3.1 Functional Areas

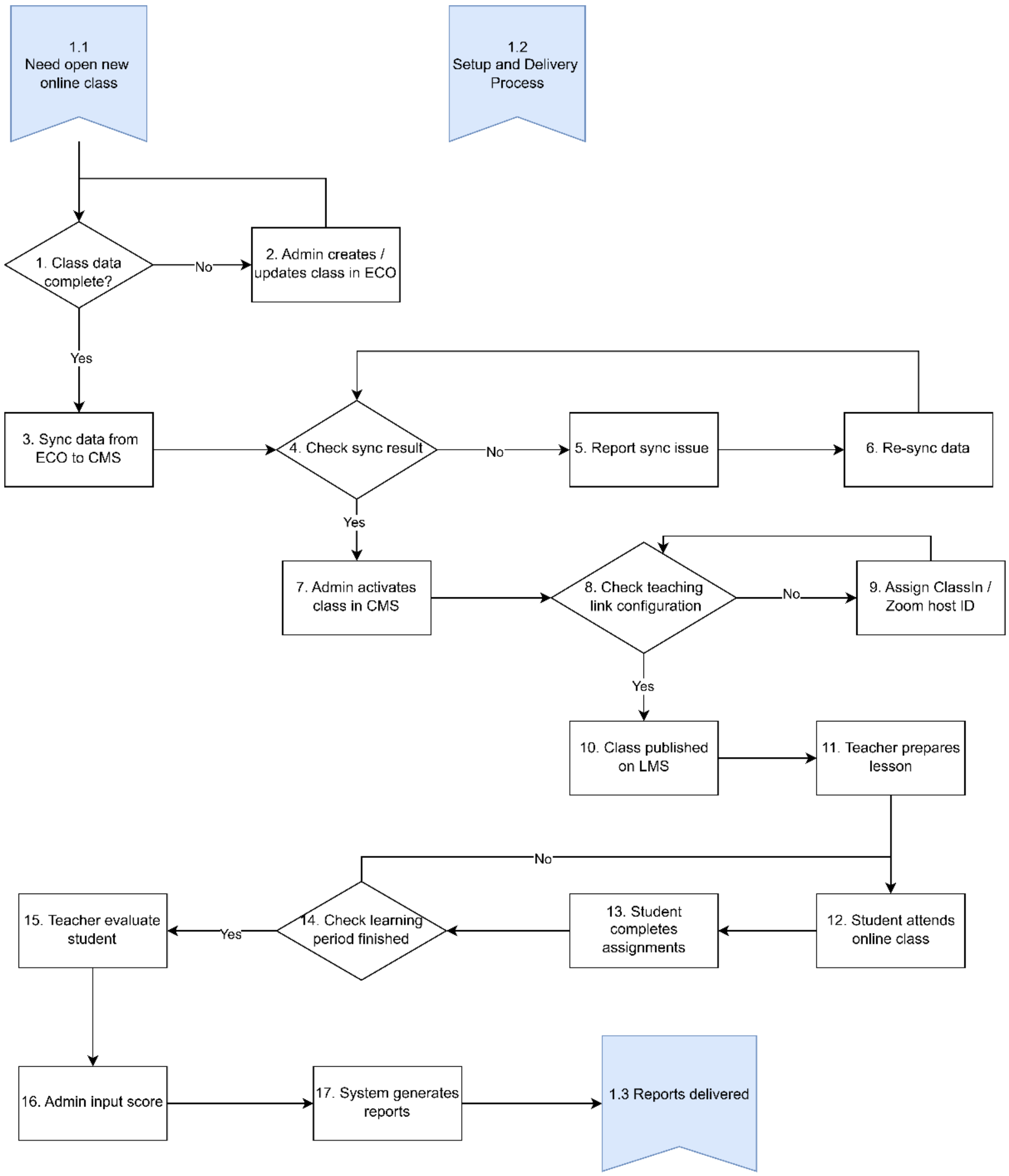
The business functions as an online learning hub, organized around a number of crucial functional domains that work together to provide top-notch instruction and student assistance:

- **Academic operations management:** Scheduling, academic reporting, instructor assignments, and class preparation and execution.
- **Teaching department:** Made up of instructors and teaching assistants who are in charge of delivering lessons, involving students, and conducting assessments.
- **Student Care Team:** Responds to questions, facilitates communication, and offers assistance to parents and students.
- **Administrative Team:** Manages student accounts, class creation, system configuration, and data entry across platforms.
- **Sales Department:** Manages customer relationships via Zalo/Facebook. When a student registers, Sales staff manually track deposits and outstanding tuition fees in their personal Excel sheets. Payment proofs (screenshots) are forwarded to a company Zalo group for verification.
- **IT Support:** Ensures platform integration, fixes system problems, and maintains technical infrastructure.

To carry out their duties, these departments rely on a variety of digital platforms (ECO, CMS, LMS, OTP2, ClassIn/ClassSeen, Zoom). However, the absence of connection across various platforms presents operational issues, requiring manual synchronization and cross-checking of data.

3.2 Core Cross-Functional Business Process: Online Class Setup and Delivery

The administrative team, academic operations, IT support, and teaching department all work together on this process. Below is a diagram and explanation of the workflow:



4 Project summary

The ILCMS (iSmart) project aims to unify the fragmented operational and financial processes of the learning center into a centralized web-based platform. Developed over 3 Agile Sprints, the system focuses on three robust pillars: Security & Role Management (Sprint 1), Academic Core setup for Courses, Classes, and Materials (Sprint 2), and Student Lifecycle & Financial Tracking (Sprint 3). By automating debt calculation, enforcing dual-verification for payments, and standardizing Excel data imports for attendance and scores, ILCMS eliminates manual cross-checking and provides a transparent, error-free operational environment for Administrators, Staff, and Sales.

5 Significance and innovation

The true innovation of ILCMS lies in shifting the center from "passive data recording" to **"Data-Driven Decision Support"**. Key innovations include:

- **Internal Control Mechanism:** A strict dual-verification payment workflow that prevents financial fraud (Sales create records, Staff verify bank transactions).
- **Data Integrity Guardrails:** Contextual locks and strict database validation during Excel bulk imports prevent corrupted academic records.
- **Strategic Dashboards:** Automated calculation of 'Class Fill Rates' and real-time Revenue/Debt tracking empowers administrators to make immediate strategic decisions on scaling up or consolidating classes.

The project develops a custom Dual-Verification workflow methodology for financial security and utilizes Contextual Data Validation algorithms to parse and authenticate bulk Excel data securely, replacing manual cross-checking.

6 Approach

The project followed an **Agile** development methodology, utilizing 3 iterative Sprints with continuous feedback from the client. The system architecture is web-based, utilizing modern frameworks (e.g., React for the frontend, Node.js/Java for the backend) and a robust centralized relational database. Strong emphasis was placed on backend constraints (e.g., restricting the deletion of active classes) and UI/UX optimization tailored to specific user roles (e.g., visual Card Views for Sales vs. Data Tables for Staff).

7 Expected outcomes

The deployment of ILCMS will yield a seamless, fraud-resistant operational flow that dramatically impacts the center's day-to-day operations. For the Sales department, the system will enable faster deal closures by providing real-time class capacity tracking, ensuring no overbooking occurs. For the Academic and Administrative Staff, the system is expected to save up to 80% of current data-entry time through automated Excel bulk imports and streamlined student profile creation. This significant reduction in manual workload will decrease human errors and free up staff to focus on student care. Most importantly, the center's leadership will gain complete, real-time visibility into financial health and academic performance via dynamic dashboards. Ultimately, the system will drive strong economic benefits by minimizing financial leakage, reducing administrative overhead, and supporting sustainable long-term growth for the institution.

8 Description of personnel

Name	Role	Responsibilities	Contributions
Nguyen Quoc Viet	Developer	Design backend & frontend; Develop core features; Integrate APIs; Optimize performance & security	Builds the technical foundation, ensuring the system is functional, secure and scalable
Vu Khanh Linh	Business Analyst	Gather requirements; Document workflows; Create user stories & diagrams; Validate business logic	Ensures the system reflects real-world needs and aligns with stakeholder expectations
Dinh Thi Hai	Project Manager	Define scope & timeline; Allocate resources; Manage communication; Monitor risks & progress	Provides strategic oversight, keeps project on tack and ensures time delivery
Nguyen Thi Hoang Yen	Tester	Design test cases; Perform functional & regression testing; Identify bugs; Validate user experience	Guarantees system reliability, usability and quality standards
Vu Lan Huong	Client Representative	Provide insights into workflows; Collaborate on requirements; Participate in testing; Act as liaison	Ensures the system meets operational needs and supports long-term growth

9 Feasibility Study

Even though the Integrated Learning Center Management System (ILCMS) hasn't been put into use yet, an initial feasibility study shows that it has a high chance of success in a number of areas.

9.1 Technical Feasibility

Highly feasible. The system relies on proven technologies (React, Relational DBs) and standard libraries for JWT authentication and Excel parsing, avoiding the complexities of heavy third-party video streaming APIs.

9.2 Operational Feasibility

Administrators, teachers, and students are already acclimated to digital platforms, which lessens the adoption hurdle. By combining disparate workflows into a single interface, the technology is designed to boost efficiency and eliminate errors. Stakeholders have expressed openness to adopt a more integrated solution, making operational feasibility promising.

9.3 Economic Feasibility

Long-term advantages include less administrative labor, cheaper maintenance expenses, and increased productivity, even if an initial investment in development and implementation will be necessary. The technology is projected to offer a good return on investment by streamlining processes and supporting scalability without commensurate increases in cost.

9.4 Social Feasibility

The system is projected to promote transparency, communication, and confidence among administrators, instructors, students, and parents. It will increase stakeholder satisfaction and improve the institution's reputation by enhancing the learning process and reporting accuracy.